flows:
dig-post-holes: ran 4 times; moved pieces_not_priced 4, post_holes_dug 4
set-floor-panels: ran 2 times; moved budget_usd -584.77, cost_usd 584.77, floor_panel_ft2_set 63.98
set-wall-panels: ran 12 times; moved budget_usd -3345.46, cost_usd 3345.46, wall_panel_ft2_set 366.02
set-roof-panels: ran 4 times; moved pieces_not_priced 1, roof_panel_ft2_set 216.01
set-posts: ran 0 times; moved
lay-subfloor: ran 0 times; moved
cut-openings: ran 0 times; moved
set-windows-and-door: ran 0 times; moved
frame-loft: ran 0 times; moved
install-boiler: ran 0 times; moved

ASSUMPTIONS, EVERY SETTING WHOSE REASON SAYS IT IS ASSUMED

floor.thickness_mm = 114 mm; reason: assumed: a 4-1/2 in (114 mm) SIP floor panel, the wall standard's thickness, research row sip.prs610_panel_thickness; Jim's floor build-up gives it
weather.u_value = 2.8 w_per_m2_k; reason: assumed: typical double glazing, not chosen by Jim
weather.solar_heat_gain_coefficient = 0.6 ratio; reason: assumed: typical double glazing, not chosen by Jim
weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side
finish.mean_reflectance = 0.5 ratio; reason: assumed mid-grey interior; Jim's wall finish gives it
finish.sky_angle_degrees = 65 degree; reason: assumed open sky; the site gives it
envelope.wall_r = 20 r_value; reason: assumed R-20 for a SIP wall (ft2 F h per BTU); the SIP thickness gives it
envelope.roof_r = 30 r_value; reason: assumed R-30; the roof build-up gives it
envelope.floor_r = 20 r_value; reason: assumed R-20; the floor build-up gives it
envelope.door_r = 5 r_value; reason: assumed R-5 for an insulated door, not checked against a source
heating.boiler_efficiency = 0.7 ratio; reason: assumed 70 percent; sources run 55 to 80 percent
joists.member_depth_mm = 140 mm; reason: assumed 2x6; the beam solver says whether it is enough
joists.spacing_mm = 406 mm; reason: 16 in on centre, assumed
header.plies = 2 count; reason: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads.roof_dead_load_psf = 15 psf; reason: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio; reason: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = south, north compass_sides; reason: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls

UNMEASURED PIECES

floor@610,2438,2438:1~2438x2438: height_mm is unknown (size-unknown)
joist@1016,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@1422,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@1828,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@2234,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@2640,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@3010,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
joist@610,2438,2438:1~38x2438: height_mm is unknown (size-unknown)
post@3048,2438,-1219: width_mm is unknown (size-unknown)
post@3048,4876,-1219: width_mm is unknown (size-unknown)
post@610,2438,-1219: width_mm is unknown (size-unknown)
post@610,4876,-1219: width_mm is unknown (size-unknown)
roof@0,0:3~3658x5486: height_mm is unknown (size-unknown)

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 2 of 2)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
7dbc993	we 073b001	not to scale	world-engine geometry; no line is drawn by hand